

# MicroDuck Arm v1-B Engineering Book

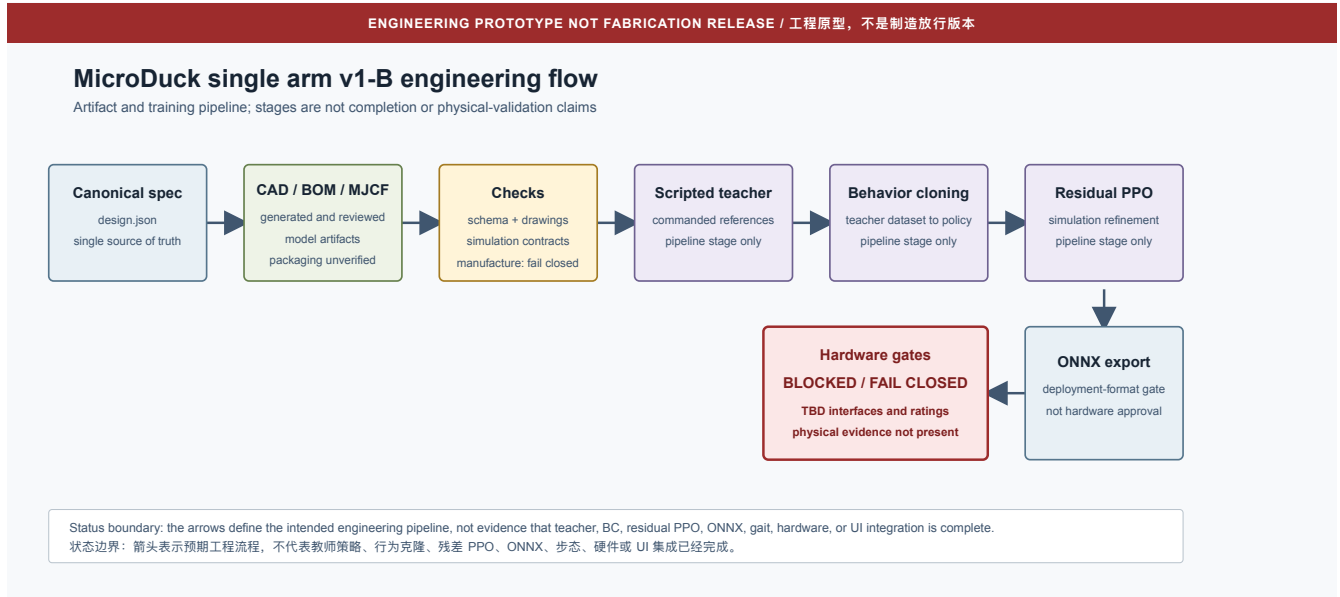


Figure 1: MicroDuck arm v1-B engineering flow

# MicroDuck single arm v1-B engineering package

## ENGINEERING PROTOTYPE NOT FABRICATION RELEASE 工程原型，不是制造放行版本

This package documents one five-actuator arm cartridge: four pose axes plus one rotary geared gripper. It provides editable OpenSCAD, review drawings, a logical electrical netlist, a bilingual BOM, a commanded test SOP, and a fail-closed acceptance record.

本文件包描述一个五执行器单臂卡匣：四个姿态轴加一个旋转齿轮夹爪。内容包括 可编辑 OpenSCAD、评审图、电气逻辑网表、双语 BOM、指令式测试 SOP 和默认 失败关闭的验收记录。

## Design boundary / 设计边界

- Nominal axes: shoulder-elbow 55 mm, elbow-wrist 50 mm, wrist-TCP 30 mm.
- Five XL330-M288-T candidates at 18 g each.
- Motor envelope 20 x 34 x 26 mm is clearance-only. The available visual mesh is about 29 mm deep and is not a manufacturing interface.
- The gripper uses two equal external gears at local  $y = +/-12$  mm, ratio -1, with 30 mm rotary jaws about parallel +Z axes. Left joint range is  $[-0.32, 0]$  rad; right is  $[0, 0.32]$  rad; right = -left.
- Central trunk mount target is 20 g; complete arm cartridge target is 160 g. Both are unverified design targets.
- Servo/horn holes, gear tooth form, bearings, fasteners, tolerances, cable exits, and base attachment are intentionally absent.
- Motor envelopes follow the canonical local body-frame centers. Gripper motor center is  $[7, 0, 22]$  mm above the hand frame; wrist motor center is  $[25, 0, 20]$  mm in the forearm frame. Overall packaging, interference, cable routing, and table clearance remain unverified.
- 标称轴距：肩肘 55 mm、肘腕 50 mm、腕部到 TCP 30 mm。
- 五颗 XL330-M288-T 候选电机，每颗 18 g。
- 20 x 34 x 26 mm 仅是间隙包络。现有可视网格深度约 29 mm，不是制造接口。
- 夹爪为两只等尺寸外啮合齿轮，本地中心位于  $y = +/-12$  mm，传动比 -1，30 mm 旋转夹指绕平行 +Z 轴运动。左关节范围  $[-0.32, 0]$  rad，右关节范围  $[0, 0.32]$  rad，并满足 right = -left；不使用理想滑块。
- 中央躯干安装件目标 20 g，完整手臂卡匣目标 160 g；两者均未验证。
- 舵机/舵盘孔、齿形、轴承、紧固件、公差、走线出口和根部连接均故意留空。
- 电机包络采用规格中的本体局部坐标。夹爪电机中心在手部坐标系  $[7, 0, 22]$  mm，腕部电机中心在前臂坐标系  $[25, 0, 20]$  mm。总体包装、干涉、走线和桌面间隙均未验证。

## Canonical source / 唯一规格源

hardware/microduck-arm-v1/spec/design.json is owned by the design lead. The generator reads it and rejects values inconsistent with v1-B. No duplicate specification is stored in this documentation slice.

hardware/microduck-arm-v1/spec/design.json 由设计负责人维护。生成器读取该文件，并拒绝与 v1-B 不一致的数值。本文件包不保存冲突的规格副本。

```
python3 hardware/microduck-arm-v1/cad/generate_engineering.py
python3 hardware/microduck-arm-v1/cad/generate_engineering.py --manufacture-check
```

The second command is expected to return nonzero until every TBD release gate has objective evidence. A zero exit would require a separately reviewed future revision.

第二条命令在所有 TBD 放行门槛获得客观证据前应返回非零。只有经过独立评审的 未来修订才可允许成功。

## Engineering and training flow / 工程与训练流程

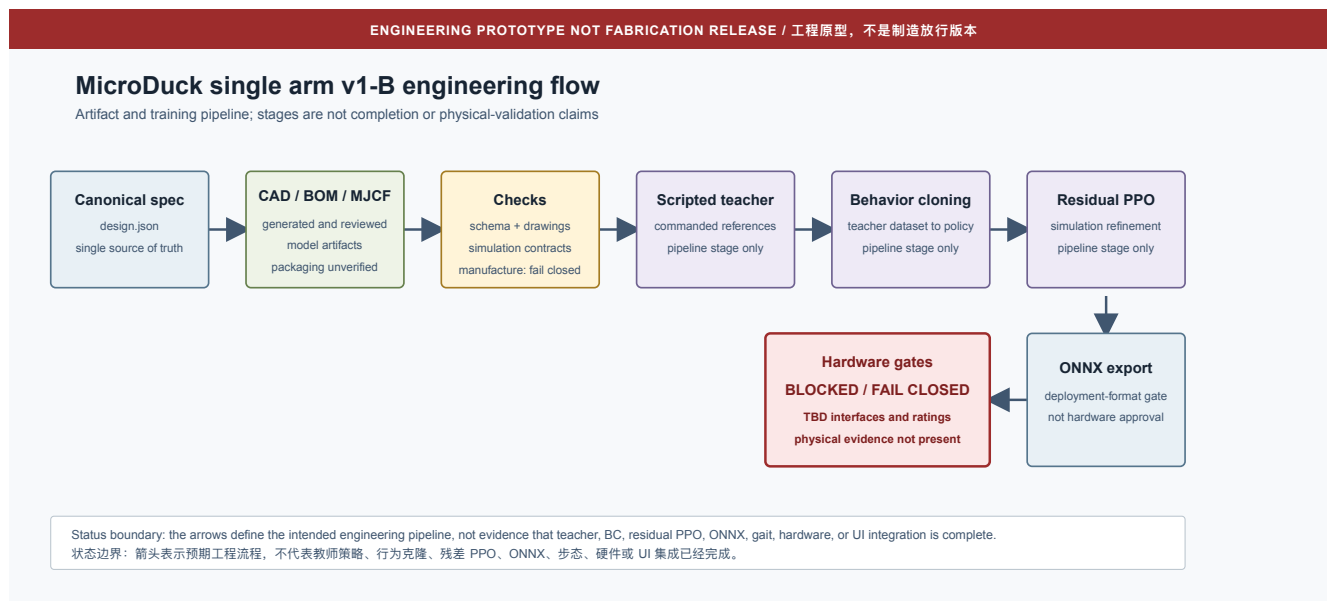


Figure 2: MicroDuck arm v1-B engineering flow

The current engineering CLI regenerates the CAD parameters, mechanical and electrical review drawings, netlist, and fail-closed manufacture record from the canonical specification:

当前工程命令行从唯一规格重新生成 CAD 参数、机械和电气评审图、网表以及失败关闭的制造检查记录：

```
python3 hardware/microduck-arm-v1/cad/generate_engineering.py
python3 hardware/microduck-arm-v1/cad/generate_engineering.py --manufacture-check
cd rlx
uv run --with pytest pytest -q tests/test_microduck_arm_v1_hardware.py
```

The intended downstream sequence is scripted teacher, behavior cloning, residual PPO, ONNX export, and then independent hardware release gates. These are pipeline stages, not claims of completed training or integration. The leader-owned [English results](#) and [Chinese results](#) are the status and evidence records when present. This package does not claim completed gait, hardware operation, or UI integration.

预期后续顺序为脚本教师、行为克隆、残差 PPO、ONNX 导出，再进入独立硬件放行门槛。这些只是流程阶段，不表示训练或集成已经完成。负责人维护的 [英文结果](#) 和 [中文结果](#) 在文件存在时作为状态和证据记录。本文件包不声明步态、硬件操作或 UI 集成已经完成。

## Build the bilingual engineering books / 构建双语工程书

```
bash docs/microduck-arm-v1/build-book.sh
```

The script requires installed pandoc, xelatex, and rsvg-convert. It fails clearly until both leader-owned result files exist, then writes:

- docs/microduck-arm-v1/MicroDuck-Arm-v1-B-Engineering.en.pdf
- docs/microduck-arm-v1/MicroDuck-Arm-v1-B-Engineering.zh-CN.pdf

脚本需要已安装的 pandoc、xelatex 和 rsvg-convert。在两份负责人结果文件都存在之前，脚本会明确失败；文件齐备后生成上述英文和中文 PDF。

## Modeling modes / 建模模式

**Fixed-root bench mode** is the only intended first physical integration: the arm root is fixed to a separately verified bench fixture, with a low workspace, catch provision, conservative software limits, and an accessible power disconnect. This package does not design or approve that fixture.

**Floating mode** is a simulation assumption only. It may use provisional mass placeholders to study system behavior, but it does not prove that a MicroDuck can carry, power, balance, or safely control this arm. No attachment to the original robot body or battery positive is approved.

**固定根部台架模式**是首个实体集成的唯一预期方式：手臂根部固定到另行验证的台架夹具，采用低位工作区、承接措施、保守的软件限位和可触及的断电装置。本文件包不设计也不批准该夹具。

**浮动模式**仅是仿真假设。可使用临时质量占位值研究系统行为，但不能证明 MicroDuck 能承载、供电、平衡或安全控制该手臂。禁止把手臂接到原机电池正极，也未批准任何机身安装。

## Files / 文件

- hardware/microduck-arm-v1/cad/microduck\_arm\_v1\_b.scad: editable concept.
- hardware/microduck-arm-v1/cad/mechanical-dimensions.svg: dimension review.
- hardware/microduck-arm-v1/cad/mechanical-dimensions.pdf: PDF export.
- hardware/microduck-arm-v1/cad/manufacture-check.json: fail-closed gates.
- hardware/microduck-arm-v1/electrical/wiring.svg: bench wiring concept.
- hardware/microduck-arm-v1/electrical/wiring.pdf: PDF export.
- hardware/microduck-arm-v1/electrical/circuit.svg: logical circuit boundary.
- hardware/microduck-arm-v1/electrical/circuit.pdf: PDF export.
- hardware/microduck-arm-v1/electrical/netlist.json: unresolved-safe netlist.
- BOM.md, SOP.md, HARDWARE-ACCEPTANCE.md: bilingual controlled records.
- engineering-flow.svg: intended pipeline with the hardware gate blocked.
- build-book.sh: English/Chinese PDF book builder; requires leader results.

Official references used for candidate identity and connector semantics:

- ROBOTIS XL330-M288-T e-Manual: <https://emanual.robotis.com/docs/en/dxl/x/xl330-m288/>
- ROBOTIS U2D2 e-Manual: <https://emanual.robotis.com/docs/en/parts/interface/u2d2/>

These references do not release the project-specific mechanical or electrical interfaces.

# MicroDuck arm v1-B BOM / 物料清单

**ENGINEERING PROTOTYPE NOT FABRICATION RELEASE**  
**No purchasing or substitution approval / 不代表采购或替代批准**

Quantities are for one arm. A named candidate is not an approved assembly until all compatibility and release gates pass.

数量按单臂计算。候选型号只有在兼容性和放行门槛全部通过后才可成为批准组件。

| ID            | Qty   | Candidate or requirement / 候选或要求                                                                                               | Status / 状态                                | Required evidence / 必需证据                                                                       |
|---------------|-------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------|
| M1-M5         | 5     | ROBOTIS XL330-M288-T, 18 g each                                                                                                | Candidate exact variant / 精确型号候选           | Purchased label, readable identity/firmware, unique bus ID, connector inspection, measured fit |
| TRUNK-BLANK   | 1     | Central trunk mount, 20 g target                                                                                               | Design target, unmeasured / 设计目标, 未测       | Released interface, weighed article, fixed-root fixture load review                            |
| ARM-CARTRIDGE | 1     | Complete cartridge, 160 g target                                                                                               | Design target, unmeasured / 设计目标, 未测       | Complete weighed assembly and center-of-mass record                                            |
| LINK-BLANKS   | 1 set | 55 mm and 50 mm axis-distance solid blanks                                                                                     | Concept only / 仅概念                         | Official mating definition, coupon fit, material/process/load review                           |
| WRIST-GRIPPER | 1     | 30 mm wrist-TCP; equal gears ratio -1; jaw centers y=+/-12 mm; 30 mm jaws; left [-0.32,0] rad, right [0,0.32] rad, right=-left | Concept only / 仅概念                         | Gear module/teeth/backlash/material, shaft support, retention, pinch review, grip calibration  |
| U2D2          | 1     | ROBOTIS U2D2, bench TTL candidate                                                                                              | Bench candidate; mobile TBD / 台架候选, 移动方案待定 | Correct TTL port, stable enumeration, communication test, mobile architecture review           |

| ID        | Qty   | Candidate or requirement / 候选或要求                     | Status / 状态                       | Required evidence / 必需证据                                                                 |
|-----------|-------|------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------|
| HUB1      | 1     | Compatible TTL/ power hub                            | Unselected / 未选型                  | Exact part, topology, current path, branch protection, connector ratings                 |
| BAT1      | 1     | Separate arm battery source for regulated 5 V domain | Unselected / 未选型                  | Chemistry, cells, capacity, fault current, enclosure, charger and transport controls     |
| BMS1      | 1     | Pack-compatible BMS/protection                       | Unselected / 未选型                  | Exact pack compatibility and voltage/current/fault/temperature ratings                   |
| F_MAIN    | 1     | Main arm feeder protection                           | Unselected / 未选型                  | Part number, rating/curve, conductor/connector coordination, measured transients         |
| F_M1-F_M5 | 5     | One protected VDD branch per actuator                | Unselected / 未选型                  | Part numbers, ratings/curves, branch fault verification                                  |
| SW_DC     | 1     | Manual DC disconnect or appropriate power pole       | Unselected / 未选型                  | DC interrupt rating at measured source fault conditions; reset behavior                  |
| HARNESS   | 1 set | GND/VDD/DATA harness with strain relief              | Unselected / 未选型                  | Pin-by-pin continuity, wire/insulation/flex ratings, polarity keying, bend/strain review |
| FIXTURE   | 1     | Fixed-root bench fixture                             | Outside released CAD / 不在放行 CAD 内 | Table/fixture dimensions, fasteners/clamp, overturning/load test, guarded workspace      |

## Provisional simulation mass assumptions / 临时仿真质量假设

The following values may be retained in a model only as **unmeasured placeholders**. They are not BOM facts, procurement specifications, measured mass, payload evidence, or physical validation:

下列数值只能作为模型中的**未测量占位值**保留。它们不是 BOM 事实、采购规格、实测质量、载荷证据或实体验证：

| Modeling placeholder / 建模占位项 | Provisional mass / 临时质量 | Provisional center (m) / 临时中心 |
|------------------------------|-------------------------|-------------------------------|
| Retained electronics / 保留电子件 | 70 g                    | [-0.010, 0, 0.018]            |
| Separate arm pack / 独立手臂电池包  | 80 g                    | [-0.040, 0, 0.015]            |
| Power hardware / 电源硬件        | 25 g                    | [-0.020, 0, 0.034]            |
| Sensor shell / 传感器外壳         | 8 g                     | [-0.020, 0, 0.059]            |

Every model and report using these values must label them `unmeasured_placeholder`. Floating-mode conclusions must be rerun after actual component selection, weighing, center-of-mass measurement, and inertia update.

所有使用这些数值的模型和报告必须标记 `unmeasured_placeholder`。实际选型、称重、重心测量和惯量更新后，必须重新执行浮动模式分析。

## Non-substitution and prohibited connections / 禁止替代与连接

- Do not substitute M077, XL320, XL430, XC330, or another voltage/protocol variant without a new design review.
- Never connect ARM\_5V to original MicroDuck battery positive or body VDD.
- Never connect arm TTL DATA to the original body actuator DATA bus.
- U2D2 is a communication candidate, not proof of servo power delivery.
- No fuse, connector, BMS, pack, or disconnect rating is inferred from a marketplace title or arithmetic stall-current sum.
- 未经新设计评审，不得以 M077、XL320、XL430、XC330 或其它电压/协议型号替代。
- 禁止将 ARM\_5V 接到原 MicroDuck 电池正极或机身 VDD。
- 禁止将手臂 TTL DATA 接到原机身执行器 DATA 总线。
- U2D2 只是通信候选，不证明能够给舵机供电。
- 不得根据网购标题或堵转电流算术和推断保险丝、连接器、BMS、电池包或断电器额定值。

# MicroDuck arm v1-B commanded test SOP / 指令式测试规程

## ENGINEERING PROTOTYPE NOT FABRICATION RELEASE

Procedure definition only; no test in this document has been run or passed.

仅定义流程；本文档不声明任何测试已执行或通过。

This SOP is for a qualified operator after component selection and an independent safety review. It gives commanded test steps, not instructions for assembling mains wiring. Use only a suitably enclosed, protected low-voltage source. Stop on unexpected motion, heating, odor, noise, current, voltage drop, communication loss, looseness, or damage.

本 SOP 供完成选型并通过独立安全评审后的合格操作人员使用。它规定指令式测试步骤，不提供市电装配说明。只使用合适的封闭式、受保护低压电源。出现意外运动、发热、异味、异响、异常电流、压降、通信中断、松动或损伤时立即停止。

## 0. Record control / 记录控制

1. Record the canonical spec revision, CAD/netlist hashes, operator, observer, date, fixture ID, source/BMS/fuse/connector part numbers, firmware, and software commit.
2. Mark every unperformed item NOT RUN; never infer a pass.
3. Keep fixed-root and floating-mode evidence separate.
4. 记录规格修订、CAD/网表哈希、操作员、观察员、日期、夹具 ID、电源/BMS/保险丝/连接器型号、固件和软件提交。
5. 未执行项目标记 NOT RUN，不得推断通过。
6. 固定根部和浮动模式证据必须分开。

## 1. Document and dead-system inspection / 文档与断电检查

1. Command no motion. Isolate the arm source and verify the documented isolation method before touching the harness.
2. Confirm exactly five M288 candidates and unique planned IDs.
3. Inspect solid support blanks and verify no one machined TBD hole patterns from the envelope or visual mesh.
4. Measure and record actual motor, horn, fastener, link, gear, jaw, cable, and fixture interfaces. Compare against released drawings only.
5. Verify pin-by-pin harness continuity and all explicit no-connects.
6. 不发送运动指令。隔离手臂电源，并在接触线束前核实文件化的隔离方法。
7. 确认正好五颗 M288 候选及计划中的唯一 ID。
8. 检查实心支撑毛坯，确认无人依据包络或可视网格加工 TBD 孔位。
9. 测量并记录电机、舵盘、紧固件、连杆、齿轮、夹指、线缆和夹具实际接口。
10. 按针脚检查线束连续性和全部明确断开点。

## 2. Power-domain command sequence / 电源域指令顺序

1. Keep all servo torque disabled in software.
2. Command the source on with current limiting/protection set by the approved electrical test plan, not by this document.
3. Record source voltage/current and each protected branch at idle.

4. Command the disconnect open; verify arm VDD is removed while noting that USB may keep U2D2 enumerated.
5. Reset only after an explicit software-safe state; reset must not command motion or automatically restore torque.
6. 软件保持全部舵机扭矩关闭。
7. 按已批准的电气测试计划设置限流/保护后上电，不使用本文档猜测额定值。
8. 记录电源和各保护支路的空闲电压、电流。
9. 指令断电器断开，验证手臂 VDD 被切断；注意 USB 可能仍使 U2D2 保持枚举。
10. 仅在明确的软件安全状态下复位；复位不得触发运动或自动恢复扭矩。

### **3. Communications and direction / 通信与方向**

1. With torque disabled, command a ping/read of one isolated motor at a time.
2. Record model identity, firmware, ID, voltage, temperature, and error state.
3. Command a very small, low-speed motion for one unloaded axis at a time in the fixed-root fixture and low workspace.
4. Verify sign, limit, zero reference, cable motion, and disconnect response.
5. For the gripper, command jaw A through a small angular increment and verify jaw B rotates oppositely through the equal-gear coupling. Reject translation or independent same-direction rotation.
6. 扭矩关闭时逐颗隔离执行 ping/读取。
7. 记录型号、固件、ID、电压、温度和错误状态。
8. 在固定根部夹具和低位工作区内，每次只对一个空载轴发送极小、低速运动指令。
9. 验证方向符号、限位、零位、线缆运动和断电响应。
10. 对夹爪，指令夹指 A 小角度转动，验证夹指 B 通过等齿轮反向转动。若出现平移或同向独立旋转则拒收。

### **4. Incremental fixed-root motion / 固定根部递增运动**

1. Command one joint at a time over a reduced range, then coordinated motion at reduced speed/acceleration.
2. At each step record commanded/observed position, tracking error, voltage, current, temperature, branch protection state, fixture motion, and cable clearance.
3. Command the disconnect during a preplanned low-energy motion and record the stop behavior; do not claim a safety category from this test.
4. Repeat only within the approved thermal/rest plan.
5. 先逐关节在缩小范围内运动，再以降低速度/加速度执行协调运动。
6. 每步记录指令/观测位置、跟踪误差、电压、电流、温度、支路保护状态、夹具位移和 线缆间隙。
7. 在预先规划的低能量运动中指令断电并记录停止行为；不得据此声明安全等级。
8. 只在已批准的温升/休息计划内重复。

## 5. Load and shoulder evidence / 载荷与肩部证据

1. Measure the complete cartridge mass and center of mass before any payload.
2. Determine worst-case shoulder gravity torque from measured mass geometry.
3. Obtain continuous-duty capability evidence for the actual actuator, thermal environment, voltage, motion profile, and support arrangement.
4. Require documented continuous capability of at least 3 x the measured worst-case gravity torque before advancing. Stall torque is not acceptable evidence.
5. Add test mass incrementally above a catch provision; the 20 g payload target remains unverified until all acceptance criteria pass.
6. 加载前测量完整卡匣质量和重心。
7. 根据实测质量几何计算肩部最坏重力矩。
8. 对实际执行器、热环境、电压、运动曲线和支撑结构取得连续工作能力证据。
9. 继续测试前，文件化连续能力必须至少为实测最坏重力矩的 3 倍。堵转力矩不能 作为证据。
10. 在承接措施上方逐步增加测试质量；20 g 载荷目标在全部验收通过前仍未验证。

## 6. Floating-mode restriction / 浮动模式限制

Do not transfer fixed-root results to floating mode. Floating simulation uses unmeasured mass placeholders and requires updated measured mass, center of mass, inertia, attachment compliance, power architecture, balance/control analysis, and a separate test authorization before any physical trial.

不得把固定根部结果直接用于浮动模式。浮动仿真使用未测量质量占位值；任何实体 试验前必须更新实测质量、重心、惯量、连接柔度、供电架构和平衡/控制分析，并取得 单独测试授权。

# MicroDuck arm v1-B hardware acceptance / 硬件验收

## ENGINEERING PROTOTYPE NOT FABRICATION RELEASE

Current state / 当前状态: NOT RUN - FAIL CLOSED / 未执行 - 失败关闭

This is an evidence template. Empty evidence means failure to release, not an implicit pass. It contains no claim that hardware exists, was operated, or passed.

本文件是证据模板。证据为空表示不能放行，不代表默认通过。本文不声明硬件已经 存在、运行或通过测试。

| Gate / 门槛                 | Required evidence / 必需证据                                                                                    | Current result / 当前结果      |
|---------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------|
| Canonical configuration   | Spec revision and generated artifact hashes match                                                           | NOT RUN                    |
| Exact actuator identity   | Five purchased XL330-M288-T units identified; firmware and unique IDs recorded                              | NOT RUN                    |
| Mechanical mating release | Official revision-controlled geometry plus physical measurements, fit coupon, tolerance review              | BLOCKED - TBD              |
| Motor depth discrepancy   | 29 mm visual mesh and 26 mm body envelope reconciled without rescaling/inference                            | BLOCKED - TBD              |
| Structural load path      | Material, fasteners, dual-side support/bearings, retention, fixture load evidence                           | BLOCKED - TBD              |
| Gripper mechanism         | Equal external gears, ratio -1, centers $y=\pm 12$ mm, 30 mm rotary jaws; tooth/backlash/retention verified | BLOCKED - TBD              |
| Packaging and clearance   | Canonical motor centers checked against real envelopes, cables, fasteners, table, fixture and full motion   | BLOCKED - UNVERIFIED       |
| Mass targets              | Trunk mount measured against 20 g target; complete cartridge measured against 160 g target                  | NOT RUN                    |
| Shoulder gravity margin   | Measured worst-case gravity torque and actual continuous-duty evidence $\geq 3\times$                       | BLOCKED - EVIDENCE MISSING |
| Separate 5 V arm domain   | Verified no connection to original MicroDuck positive/body VDD/body DATA/USB VBUS                           | NOT RUN                    |
| Pack and BMS              | Exact compatible parts, ratings, enclosure, charger and fault review                                        | BLOCKED - TBD              |
| Protection                | Main and five branch protective devices coordinated with measured loads and conductors                      | BLOCKED - TBD              |
| Harness/connectors        | Pinout, continuity, polarity keying, ratings, strain relief, flex and routing evidence                      | BLOCKED - TBD              |
| Disconnect behavior       | Correct DC-rated device; open/reset behavior tested without automatic motion                                | BLOCKED - TBD              |

| Gate / 门槛                | Required evidence / 必需证据                                                              | Current result / 当前结果 |
|--------------------------|---------------------------------------------------------------------------------------|-----------------------|
| U2D2/hub bench interface | Stable communication and verified topology; no assumption that U2D2 powers motors     | NOT RUN               |
| Mobile interface         | Compute, communication, power and mounting architecture                               | BLOCKED - TBD         |
| Fixed-root fixture       | Geometry, fasteners/clamp, overturning/load, workspace and catch review               | BLOCKED - OUTSIDE CAD |
| Software limits          | Joint signs, zeros, ranges, speed/acceleration/current/temperature responses recorded | NOT RUN               |
| Incremental motion       | Unloaded and reduced-range commanded tests completed per SOP                          | NOT RUN               |
| Payload target           | Incremental test through 20 g with thermal/tracking/grip/drop evidence                | NOT RUN               |
| Floating mode            | Measured mass/inertia/attachment/power/balance evidence and separate authorization    | BLOCKED               |

## Acceptance rule / 验收规则

Release requires every applicable row to contain traceable evidence and a reviewed PASS. NOT RUN, TBD, BLOCKED, missing data, estimates, simulation only, or fixed-root evidence reused for floating mode all fail closed.

放行要求每个适用条目都包含可追溯证据并经评审标记为 PASS。NOT RUN、TBD、BLOCKED、缺失数据、估算、仅仿真，或把固定根部证据复用于浮动模式，均按失败关闭。

## Sign-off / 签署

| Role / 角色                  | Name / 姓名 | Revision / 修订 | Date / 日期 | Decision / 决定 |
|----------------------------|-----------|---------------|-----------|---------------|
| Mechanical reviewer / 机械评审 |           |               |           | NOT REVIEWED  |
| Electrical reviewer / 电气评审 |           |               |           | NOT REVIEWED  |
| Controls reviewer / 控制评审   |           |               |           | NOT REVIEWED  |
| Test lead / 测试负责人          |           |               |           | NOT REVIEWED  |

# MicroDuck Single-Arm v1-B — implementation and evidence

## Decision: prototype delivered; integrated robot release blocked

This is an executable **engineering-validation prototype**, not a fabrication-ready robot or a claim that every task passed. The current reference evaluation is artifacts/microduck-arm-v1/evaluation-attempt-3/evaluation.json. Attempts 1 and 2 are historical, with older environment hashes; do not combine their results with the current revision.

Software tests, simulated task success, artifact integrity, and physical acceptance are four different claims. In particular, a fixed-root reach success does not establish walking, carrying, balance, electrical safety, or hardware readiness.

Final targeted software run: **137 passed** (42 new-package tests and 95 previous-arm regression tests), with seven ONNX-export/tracing warnings. Evidence: artifacts/microduck-arm-v1/software-tests.log and software-tests.xml. This does not mean the physical task suite passed.

## Implemented package

| Deliverable                                     | Repository location                                              | Qualification                                                           |
|-------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------|
| Canonical dimensions, masses and joint contract | hardware/microduck-arm-v1/spec/design.json                       | Assumptions and unknowns are explicit                                   |
| BOM and assembly/test procedure                 | docs/microduck-arm-v1/BOM.md, SOP.md                             | Exact motor candidate; other procurement interfaces remain TBD          |
| Editable CAD and dimension drawings             | hardware/microduck-arm-v1/cad/                                   | SCAD, SVG and PDF review geometry; no released mating holes or STEP/STL |
| Circuit, wiring and netlist                     | hardware/microduck-arm-v1/electrical/                            | Logical topology; not a fabrication-ready PCB or qualified harness      |
| Integrated dynamic model                        | hardware/microduck-arm-v1/generated/microduck-arm-v1-free.xml    | Free robot root; physical task-object contacts                          |
| Explicit bench model                            | hardware/microduck-arm-v1/generated/microduck-arm-v1-fixture.xml | Fixed-root assistance, never a mobile validation substitute             |
| URDF                                            | hardware/microduck-arm-v1/generated/microduck-arm-v1.urdf        | Kinematic interchange with documented backlash/collision approximations |
| Model, workspace, torque and task validation    | microduck_arm_v1/validation.py                                   | Executed; failures preserved                                            |
| BC, residual PPO, ONNX and evaluation           | microduck_arm_v1/learning.py                                     | Simulation training tools, not an onboard controller                    |
| Evidence audit and video gallery                | microduck_arm_v1/evidence.py                                     | Hash, trace and video-frame integrity checks                            |
| Hardware acceptance                             | docs/microduck-arm-v1/HARDWARE-ACCEPTANCE.md                     | Not run; missing evidence blocks release                                |

## Design consistency and deliberate corrections

- **Four pose axes plus one gripper = five actuators**, not “5 pose DOF plus gripper.” Ten retained leg actuators plus five arm actuators give a new **15-action / 64-observation** contract. Existing 14-action / 61-observation policies must not be loaded into this model.

- v1-B uses five identical XL330-M288-T candidates. This retains one motor identity and interface definition throughout the new package. It does not prove torque adequacy or imply that an M077 substitution was validated.
- Nominal axis distances are **55 / 50 / 30 mm**, totaling 135 mm. This geometric sum is not a certified payload reach or balanced workspace.
- The gripper uses two rotary fingers and a -1 gear coupling. MJCF equality and URDF mimic describe this proposed transmission; real gears, backlash, bearings and strength still need engineering.
- Motor envelopes are 20 × 26 × 34 mm in the model' s width/depth/height convention. The old visual mesh depth differs; neither visual geometry nor a bounding box supplies trustworthy mounting-hole coordinates.
- The original model' s removed neck subtree weighs 0.2799274 kg, not merely an empty head shell. Electronics, battery, power hardware and sensor shell are therefore explicitly reintroduced as **unmeasured mass assumptions**, rather than silently deleted.
- The resulting modeled robot mass is **0.80631578 kg**, excluding the task object. This is not a weighed article. No mass/CoM claim is hardware-qualified.
- The arm positive power domain is kept separate from the body positive domain. Final pack/BMS/DC-DC/fuse/connector selections, regenerative-energy protection and thermal tests remain unresolved. No real actuation or onboard software change was performed.

## Actual simulation results

Teacher: IK plus a time-phased state machine. Seeds: 910000 and 910001. Limit: 1,500 control steps, 0.02 s per step; episodes end early on success or failure. Payload in the task suite: 10 g. It is a modeled task object, not a rated payload.

| New case                  | Free-base result | Fixture result | Observed limitation                                     |
|---------------------------|------------------|----------------|---------------------------------------------------------|
| Reach                     | 0/2              | 2/2            | Free-base fall                                          |
| Pick/place                | 0/2              | 0/2            | Environment contact / robot self-collision              |
| Obstacle relocation       | 0/2              | 0/2            | Environment contact / robot self-collision              |
| Stationary waypoint carry | 0/2              | 0/2            | Environment contact / robot self-collision              |
| Stowed-arm walking        | 0/2              | Not allowed    | No working gait teacher; environment contact            |
| Walking while carrying    | 0/2              | Not allowed    | Environment contact; no validated whole-body controller |
| <b>Total</b>              | <b>0/12</b>      | <b>2/8</b>     | <b>Integrated task release fails</b>                    |

The two fixture reach episodes end with TCP errors of **4.63 mm and 4.09 mm**, and full rotation errors of **0.0457 and 0.0427 rad**. The acceptance window also requires a one-second hold. These are two seeded simulation outcomes, not an accuracy guarantee or a 40/40 campaign.

The previous lab' s single-arm reach/pick/relocate/carry tasks have corresponding new scenarios, but their old pass records cannot transfer to a different morphology. arms-handover-v1 and arms-

co-carry-v1 require phase two and are not implemented on a single arm. Painting, DAgger, color coverage/precision, and real contact-force qualification are also not implemented here.

## Acceptance protections

Forbidden contacts are latched across physics substeps. Reach checks a full target rotation, not just one tool axis. Grasp loss after a valid lift is failure unless release is authorized in a valid placement phase; authorization is revocable and unavailable to walking/carrying. Walking requires recent ordered alternating support events while upright and making forward progress. Stowed walking checks arm posture; carrying checks current lift, bilateral grasp, absence of table/floor support, and a bounded body-relative object pose.

The default 30 s horizon includes the teacher's 21 s release phase and one-second hold. Shorter runs may be deliberate smoke tests, not complete manipulation evaluations. The current gait teacher merely commands the home pose; it is not a locomotion implementation.

## Workspace and torque screening

- 10,000 seeded target samples in the specified Cartesian box: **3,331** meet the constrained horizontal-tool IK test; **2,831** additionally pass the collision screen.
- Balance-safe count is **unknown**. These counts are not a usable-workspace rating and do not establish arbitrary six-dimensional reachability.
- A 500-pose quasistatic table with an assumed 20 g TCP payload gives a maximum required qualified continuous torque at the  $3\times$  screening factor of **0.33063 N · m**. The table includes infeasible poses, explicitly flagged.
- Actual qualified continuous motor capability is unknown. The model's  $\pm 0.10$  N · m actuator limit is a screening assumption, not a manufacturer continuous-duty rating. Dynamic acceleration, contact loading, heating and battery behavior are not qualified.

Raw files: artifacts/microduck-arm-v1/workspace.json and artifacts/microduck-arm-v1/mechanics/torque-com.csv.

## Training and video evidence

The learning smoke uses successful fixture reach demonstrations, BC, residual PPO with an external IK teacher, and held-out evaluation. Exact checkpoint paths, losses, returns, normalization, source hashes and ONNX parity are in artifacts/microduck-arm-v1/learning-smoke/. Consult the final learning summary rather than historical attempts. A short training smoke is not convergence, a successful mobile policy, or proof that PPO improves the teacher.

The actor currently receives simulator-derived TCP/object/goal poses. Hardware state estimation is absent. Residual ONNX output requires the matching teacher, observation normalization and action mapping; it is not a drop-in Radxa or robotd policy.

## Recorded learning metrics

Seed 101 produced eight successful demonstration episodes and 643 state/action samples. BC ran 20 epochs; residual PPO ran 2,048 environment steps and 32 optimizer updates. The independent evaluation seeds were 201 - 205, with an explicit 250-step horizon for this short reach smoke.

| Controller             | Held-out success | Mean return | Mean steps |
|------------------------|------------------|-------------|------------|
| IK/FSM teacher         | 5/5              | -0.033414   | 88.0       |
| BC                     | 0/5              | -10.173485  | 250.0      |
| Teacher + residual PPO | 5/5              | -0.020445   | 87.6       |

BC action MSE fell from 0.097145 to 0.00033247, **but every held-out BC episode timed out.** PPO's first/last logged training loss was 0.090142/0.048559; these totals need not decrease monotonically. Its 23 completed training episodes had mean return -0.102520. Raw JSONL also records value loss, policy-gradient loss, entropy, approximate KL, clipping and explained variance. Five small paired evaluations do not establish a statistically reliable PPO improvement over the teacher.

ONNX numerical parity passed on separate seeds 301 - 305: BC maximum absolute error 1.55e-7, raw PPO residual 2.24e-8. The residual is not a standalone action policy.

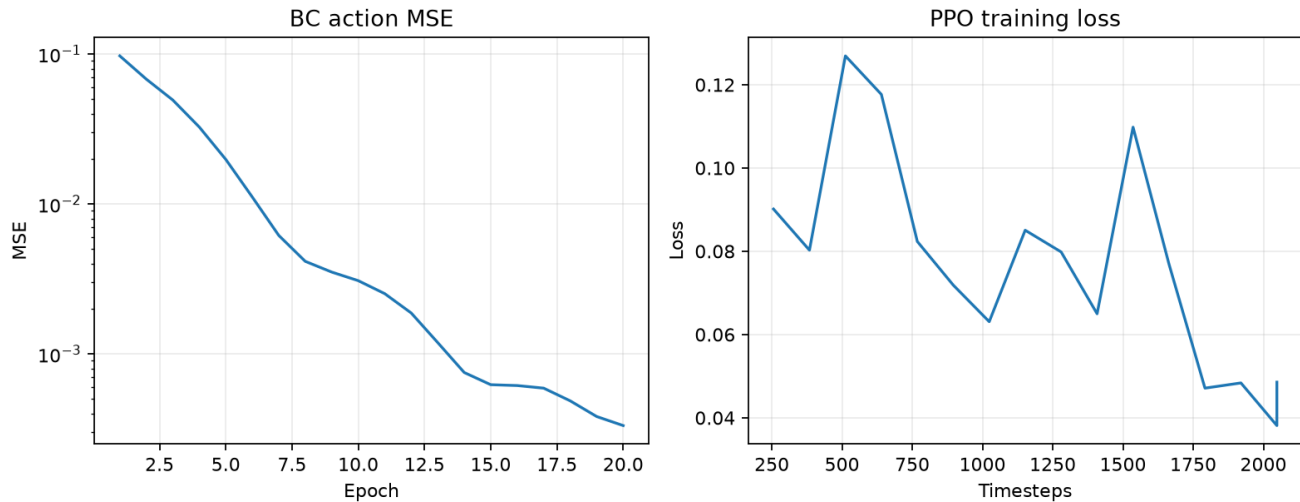


Figure 3: Actual BC and PPO loss histories

Ten current IK/FSM MP4s cover all six free-base and four fixture cases, including failures. Open artifacts/microduck-arm-v1/verified-evidence/videos.html. Its index checks each MP4 hash, H.264/25fps encoding, terminal-inclusive sampled-frame count and the associated episode trace. The terminal frames of fixture reach and free reach were visually inspected: fixture success and free-base fall are visible. Video integrity is not task success.

The same gallery also includes two actual post-training checkpoint videos: held-out seed 201, BC time-out and teacher-plus-residual-PPO success. **Twelve current videos total**, with trace and checkpoint hashes. The checkpoint videos are under artifacts/microduck-arm-v1/learned-videos/; they do not reuse teacher footage as learned-policy evidence.

No-actuation and open-gripper controls are stored in artifacts/microduck-arm-v1/negative-controls-final/. Open gripper is not a meaningful negative for reach. When the teacher itself fails a manipulation case, control failures cannot establish a valid manipulation benchmark.

## Reproduction

Run from the workspace root using the existing environment; no dependency installation is required in this checkout. Tested runtime versions: MuJoCo 3.10.0, Stable Baselines3 2.9.0, PyTorch 2.9.1 and ONNX Runtime 1.29.0. PDF tools and ffmpeg/ffprobe are separate system tools. OpenSCAD CLI is not installed here, so no fabrication mesh export was verified.

```
export PYTHONPATH=.:rlx
PY=rlx/.venv-microduck/bin/python
$PY -m microduck_arm_v1 build
$PY -m microduck_arm_v1 check
$PY -m microduck_arm_v1 torque --out artifacts/my-v1/torque --samples 500
$PY -m microduck_arm_v1 workspace --out artifacts/my-v1/workspace.json --samples 10000
$PY -m microduck_arm_v1 evaluate --out artifacts/my-v1/eval --episodes 2 --max-steps 1500 --videos
$PY -m microduck_arm_v1.evidence controls --out artifacts/my-v1/controls
```

```
$PY -m microduck_arm_v1.evidence audit --evaluation artifacts/my-v1/eval/evaluation.json --out artifacts/my-v1,  
$PY -m microduck_arm_v1 demonstrate --case reach --mode fixture --episodes 8 --max-steps 250 --out artifacts
```

Use the dataset path printed by demonstrate with train-bc --dataset ... --epochs 20 --out .... Use train-ppo --case reach --mode fixture --steps 2048 --max-steps 250 --out ... for the small residual smoke. Use the printed checkpoint with export --checkpoint ... --out .... The API learning.evaluate\_policy requires explicit held-out seeds. The final smoke script and result artifacts supply an exact runnable example.

evaluate intentionally exits **2** when free-base tasks fail. hardware-check also exits **2** without required measured evidence. generate\_engineering.py --manufacture-check exits **3** while fabrication requirements are unresolved. Do not mask those statuses as passes. On macOS, offscreen MuJoCo may require permission to access graphical services.

## Remaining work before physical integration

1. Measure the actual duck, motors, mating interfaces, feet, battery and electronics; replace mass/inertia placeholders and resolve the motor-mesh discrepancy.
2. Revise the load-bearing mount and collision-free arm motion; current pick trajectories collide with the robot. Do not disable collisions to manufacture success.
3. Develop and validate a whole-body standing controller before locomotion; then implement a gait and repeat loaded free-base evaluations.
4. Complete actual mating CAD, tolerances, transmission and structural calculations; select and qualify electrical protection, energy storage and harnesses. The current logical diagrams cannot authorize fabrication.
5. Add sensor estimation, timing qualification, safe robotd/control-center integration and independent hardware acceptance. The viewer's earlier arm page does not establish support for this new 64/15 contract.
6. Retrain and compare teacher/BC/PPO across preregistered task and perturbation conditions. Only then advance through restrained bench, standing, contact and slow-carry hardware tests.

The deliverable stops at reproducible prototype artifacts and explicit failure evidence. **No full-skill pass, physical capability rating, fabrication release, or hardware safety certification is asserted.**